

DESIGN AND IMPLEMENTATION OF AN IN-VACUUM MAGNETIC FIELD MEASUREMENT SYSTEM FOR THE TPS NONLINEAR IN-VACUUM KICKER

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Abstract

The Taiwan Photon Source (TPS) storage ring utilizes a conventional four-kicker bump off-axis injection scheme, which may induce beam perturbations during the injection process. To enhance beam stability and enable reliable top-up operation, a novel Nonlinear In-Vacuum Kicker has been developed. This device is engineered to produce zero magnetic field components in both the horizontal (B_x) and vertical (B_y) directions at the beam center, while generating a peak B_y field at the injection point. This magnetic field configuration enables transparent injection with minimal disruption to the stored beam.

To precisely characterize the magnetic field distribution of the NIK and eliminate arcing issues encountered during atmospheric measurements, a vacuum-compatible magnetic field measurement system has been designed and implemented. The system enables accurate field mapping across various excitation currents. This paper presents the system's mechanical design, fabrication process, and initial experimental validation results.

INTRODUCTION

The National Synchrotron Radiation Research Center (NSRRC) is advancing an innovative injection scheme for the TPS that incorporates a nonlinear injection kicker. Unlike conventional implementations used at facilities such as MAX IV, BESSY II, and Sirius which operate out-of-vacuum, the TPS adopts a novel in-vacuum nonlinear injection kicker design [1]. Consequently, we have named it the Nonlinear In-vacuum Kicker (NIK). This configuration is expected to improve injection performance while minimizing interference with the stored beam.

However, during initial magnetic field measurements under atmospheric conditions, electrical arcing was observed due to the system's high-voltage characteristics. To address this issue and improve operational safety and reliability, a vacuum-compatible magnetic field measurement system was developed. The system was designed based on the NIK magnet structure, specifically optimized to prevent electrical discharge in high-voltage environments. The magnet features a magnetic length of 0.36 m and a racetrack-shaped aperture of 68 mm (horizontal) by 8 mm (vertical). With a nominal current of 1588 A, the peak field reaches 895 Gauss, yielding an integrated field of 32,127 Gauss · cm. The printed circuit board (PCB) coil

used for signal detection has a total length of 0.75 m, allowing high-resolution measurement of the magnetic field profile.

MECHANICAL DESIGN

To facilitate the development of the NIK at TPS, a dedicated magnetic field measurement system was designed and constructed. This system utilizes a PCB coil to measure the integrated magnetic field, as illustrated in Fig. 1. Optimized for vacuum operation, it features high-precision PCB translation capabilities, allowing for a motion range of ± 20 mm along the X-axis and ± 2 mm along the Y-axis. The motion stages are equipped with precision actuators, enabling a positioning resolution of up to 10 μ m. These specifications fulfill the spatial resolution requirements essential for accurate magnetic field profiling, making the system well-suited for evaluating complex nonlinear magnetic field structures.

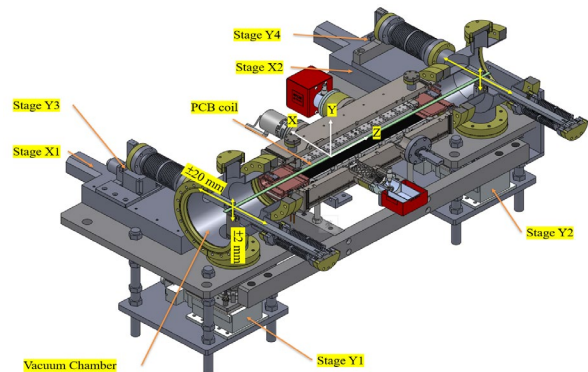


Figure 1: Vacuum magnetic field measurement system with PCB coil on X–Y stages for the NIK at TPS.

In designing the vacuum chamber, special attention was given to evaluating the mechanical stability and positional behaviour of the structure under both atmospheric and vacuum conditions, ensuring consistent operational reliability and motion accuracy across varying pressure environments.

During system operation, vacuum-induced compression and tension in the bellows generate non-uniform reaction forces on the motion platform Y3 and Y4, particularly during transitions between these two states. This mechanical imbalance leads to platform instability, causing the PCB coil to experience undesirable vibrations or sudden jumps a phenomenon especially noticeable when the PCB is positioned near X = 0 mm, where the platform is most sensitive to force fluctuations.

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To counteract this effect, a spring-loaded bolt mechanism (Fig. 2(a)) was implemented between the vacuum chamber and the displacement platform. This structure applies a stable preload to the underlying motion stage, enhancing contact stability and effectively absorbing dynamic forces from the bellows, thereby resolving vibration and jumping issues.

In addition, when the PCB moves along the X-axis, any misalignment or non-parallel motion between its two supporting ends can result in deflection and sagging of the board, which further compromises its flatness and the accuracy of magnetic field measurements. To address this issue, a spring-tension mechanism was introduced (Fig. 2(b)). A pair of preload springs applies continuous longitudinal force at both ends of the PCB, especially providing compensation during non-parallel motion along the X-axis, effectively preventing midspan sagging, maintaining PCB flatness, and ensuring scanning precision.

These mechanical enhancements including the spring-loaded bolts and tension springs significantly improve the dynamic stability of the motion platform and vacuum interface, thereby ensuring the reliability and precision of in-vacuum magnetic field profiling.

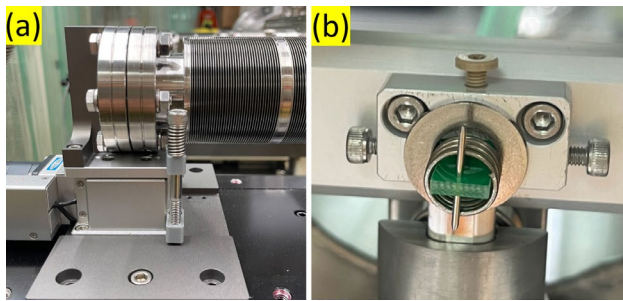


Figure 2: (a) Spring-loaded bolt applying preload to the displacement stage. (b) A pair of preload springs applies tensile force to the PCB during non-parallel movement in the X direction to maintain its flatness.

FIELD PERFORMANCE

To support high-precision magnetic field profiling, a dedicated measurement system was developed, as illustrated in Fig. 3. The setup consists of a high-voltage pulsed power supply driving a NIK magnet, which induces signals in a long sensing coil. These signals are processed through an RC integrator, while a current transformer (CT) captures the pulse current. All signals including the induced voltage (CV), current, and trigger are synchronized and recorded using a function generator and oscilloscope. This configuration ensures accurate detection and time-aligned acquisition of transient magnetic responses.

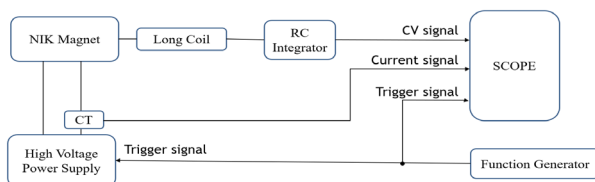


Figure 3: Schematic of the NIK magnetic field measurement system synchronized by a function generator.

To assess the system's stability and repeatability under vacuum conditions, a series of magnetic field scans was conducted using identical operational parameters. At each sampling point along the scan path, the magnetic signal was recorded 32 times and then averaged to suppress noise and enhance measurement consistency. As shown in Fig. 4, the four measurements of the first integral of the vertical magnetic field component display highly consistent profiles across the full X-axis range. Notably, at the injection point ($x = -15$ mm), the system achieves an impressive repeatability of $\pm 0.33\%$, underscoring its excellent performance. These results validate the mechanical rigidity, precise motion control, and reliable signal acquisition of the setup, confirming its suitability for high-precision magnetic field characterization.

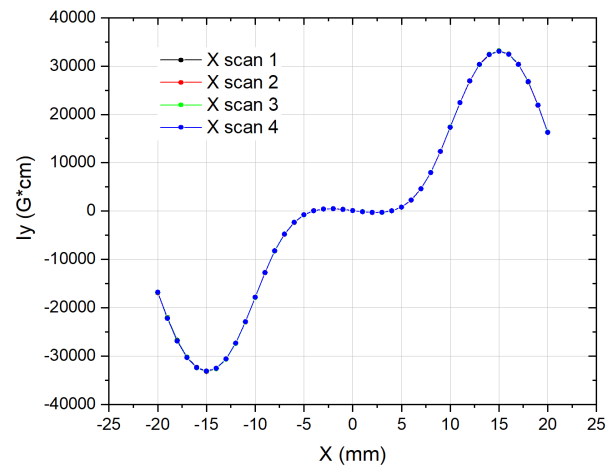


Figure 4: Excellent repeatability across four x-axis scans confirms system stability.

To further assess the accuracy and applicability of the developed in-vacuum magnetic field measurement system, experimental results were compared with simulation data under realistic operating conditions relevant to the NIK at TPS. The system employs a PCB coil to scan the magnetic field along both the X and Y axes, with particular attention given to alignment accuracy and positional calibration to ensure meaningful and accurate comparisons.

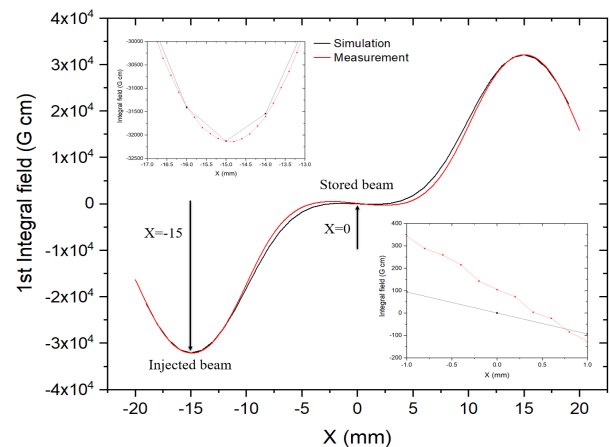


Figure 5: Good agreement between simulation and measurement, confirming field alignment and distribution.

Figure 5 shows a comparison of the first integral of the vertical magnetic field component (B_y) along the X-axis between the measured data (red) and the simulation (black). The zero-integral region at $x=0$ mm corresponds to the free-field zone where the stored beam passes, minimizing magnetic disturbance during top-up injection. Meanwhile, the injected beam traverses the kicker at $x=-15$ mm, aligning with the magnetic field peak to ensure sufficient kick strength. The insets highlight close agreement near these two key locations.

Figure 6 presents the first integral of the horizontal field component (B_x) at different Y-positions of the PCB coil: -0.20 mm (black), -0.26 mm (red), and -0.30 mm (green). The $Y = -0.26$ mm position yields the most symmetric and centered field profile, matching the design expectation and optimizing the measurement region sampled by the PCB. This confirms the importance of precise vertical alignment for accurate magnetic field characterization.

Theoretically, the optimal measurement position should be located at $Y = 0$ mm, where the PCB coil aligns with the magnetic mid-plane, assuming a perfectly symmetric magnetic field design. At this position, the first integral of the field is expected to exhibit maximum symmetry and match the intended field profile. However, as shown in Fig. 5, the most symmetric result was observed at $Y = -0.26$ mm, indicating a shift in the effective magnetic center. This discrepancy may be attributed to installation errors, mechanical misalignments, or slight asymmetries in the actual magnet structure.

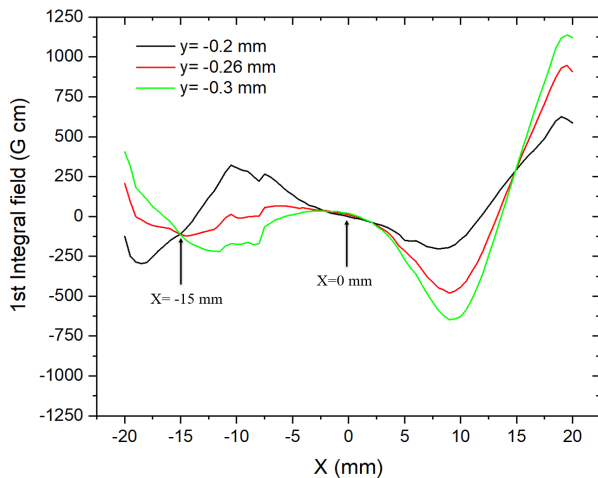


Figure 6: At $Y = -0.26$ mm, B_x integral is smallest near $x = 0$ mm and $x = -15$ mm, and flattest between $x = -15$ mm and 0 mm, reducing beam disturbance.

To investigate this deviation, a 3D coordinate measuring machine (CMM) was employed to perform detailed dimensional measurements of the mechanical structure. As shown in Fig. 7, the measurement results revealed a vertical offset between the PCB scanning plane and the designed magnetic mid-plane. This high-resolution geometric data helped identify the source of misalignment and confirm the physical origin of the observed shift. The find-

ings have been incorporated into the analysis and calibration process to improve alignment precision and enhance the overall performance of the measurement system.

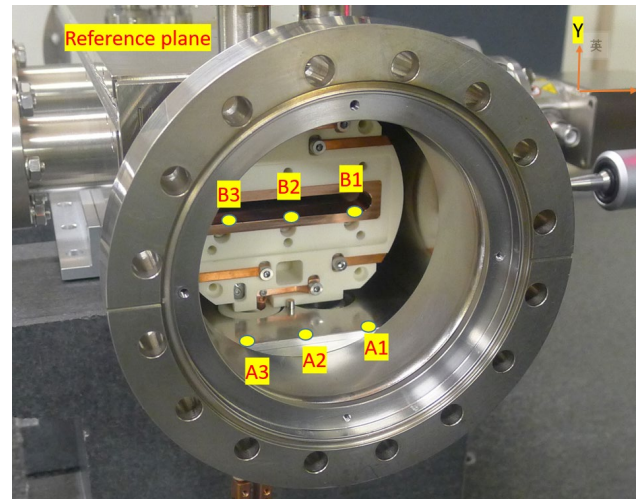


Figure 7: Internal view of the chamber with probe positions. $A1 \rightarrow A3$ and $B1 \rightarrow B3$ are reference points on the lower and upper electrodes, the Reference plane is used as the CMM alignment datum.

After referencing the coordinate system to the reference plane, three points were measured on each of the A and B surfaces and averaged. The result showed a vertical offset of $Y = -0.283$ mm from the design reference surface. This value is consistent with the magnetic measurement system result of $Y = -0.26$ mm, indicating good agreement between the mechanical alignment and the magnetic field center, and confirming the correlation between physical geometry and magnetic field distribution.

SUMMARY

The successful implementation of the vacuum-compatible magnetic field measurement system effectively resolved the high-voltage arcing issues previously encountered during atmospheric measurements. Moreover, the measured magnetic field data exhibit strong agreement with simulation results, confirming both the functionality and reliability of the system.

A 0.26 mm offset between the magnetic field center and the mechanical center, observed in the B_x measurement, was further verified using a coordinate measuring machine. This deviation is attributed to cumulative mechanical tolerances introduced during component assembly. To enhance alignment precision in future iterations, refinement of the assembly and alignment procedures will be considered.

REFERENCES

- [1] C. C. Chang *et al.*, "Development of the Nonlinear In-Vacuum Kicker at Taiwan Photon Source", *Vacuum*, vol. 239, p. 114356, 2025. doi:10.1016/j.vacuum.2025.114356